NutriMed Ask — Product Specification

Irish Nutrition and Dietetic Institute (INDI)

1. Product Overview

NutriMed Ask is a nutrition and medical information assistant built for the Irish Nutrition and Dietetic Institute (INDI). It helps members of the public with medical conditions understand their diagnosis, dietary requirements, treatment options, and how to self-manage their health confidently, with responses grounded exclusively in INDI-approved, HSE-validated resources.

2. User Persona

John Connolly, 50, was recently diagnosed with Type 2 diabetes following years of sedentary work and poor diet as an accountant. He needs to keep his blood glucose within safe limits to avoid complications affecting his eyes, kidneys, and cardiovascular system.

3. Problem: How John Gets Information Today

John wants to understand hypoglycemia and know what to do when his glucose drops below 4 mmol/l. Today his journey looks like this:

- His GP points him to the INDI website. He navigates to the diabetes section and finds nine PDF links, none labelled clearly enough to tell him which covers hypoglycemia.
- He opens and scrolls through each PDF, gives up after a few minutes, and falls back to a Google search. The results are a mix of sources with no clear indication of which are trustworthy.
- He tries ChatGPT as a shortcut. The answers are long, generic, and not tailored to his situation. He ends up down rabbit holes of follow-up questions unrelated to his original query.

4. Why RAG?

The solution needs to be grounded in science-backed, dietitian-approved content from INDI specifically. A general-purpose LLM with better prompting is not enough for a medical context where accuracy and source provenance are non-negotiable.

- Web search returns results from unverified, mixed sources.
- INDI already publishes all the required content on its website and PDFs, thoroughly reviewed and approved by HSE Ireland.
- RAG pins every answer to that trusted content, making the system explainable and auditable.

5. User Stories

#	Use Case	User Story	Scope	Priority
1	Answer simple queries directly from an INDI PDF	As a user with diabetes, I want to know how many carbs a Type 1 diabetic should eat per meal, so that I can follow a plan to manage my glucose levels.	In Scope	P0
2	Answer complex queries requiring reasoning across more than one INDI PDF	As a user with Type 1 diabetes on insulin, my blood ketone is 0.9 mmol/l and urine ketone is positive — should I retest glucose and ketones, so I can know if I need extra insulin?	In Scope	P0
3	Answer queries with measurable quantities (glucose levels, carb grams, fluid volumes)	As a user with diabetes, when my blood glucose is below 4 mmol/l, how many grams of carb should I consume?	In Scope	P0
4	Queries requiring human intervention — clinical complexity beyond INDI scope	As a user with cardiac risks on blood pressure medication, my LDL cholesterol is above 4 mmol/l — should I cut dairy, red meat, and go vegan, so I can bring my cholesterol back to normal?	In Scope — reassure with validating message, hand over to human experts	P1
5	Emergency queries — refuse and escalate immediately	I am experiencing breathlessness, lack of energy, and extreme thirst due to high blood glucose. Can you suggest any OTC medication I can get from a pharmacy immediately?	In Scope — alert user with emergency box, notify human experts, show 'connecting to a team member' status	P1
6	Queries that cannot be answered from INDI data sources	As a user with a child showing muscular weakness symptoms, should I look for genetic testing to confirm muscular dystrophy?	Out of Scope — this condition is not in INDI's data source; politely decline to answer	P2

6. Queries the Chatbot Refuses or Escalates

Sensitive queries — refuse and escalate immediately

Example: "I have had fatigue and extreme thirst for several weeks. Can you suggest an over-the-counter medication for diabetes?"

- Suggesting medication for potentially undiagnosed conditions falls outside the chatbot's remit.
- Do not pass to the LLM. Escalate to a clinical staff member immediately.

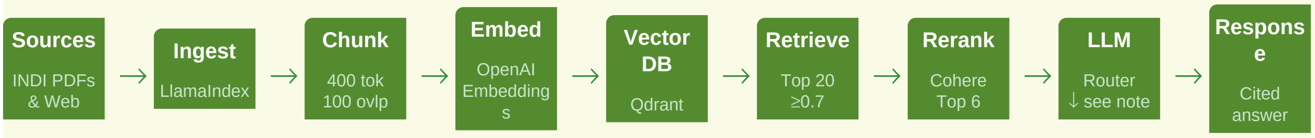
Out-of-scope queries — decline gracefully

Example: "Can you recommend the best private hospital or dietitian for diabetic care in Ireland?"

- Third-party recommendations are outside INDI's scope. The chatbot declines and suggests the user contact INDI directly.

7. System Architecture

The pipeline below shows the full data flow from INDI source documents to a cited response delivered to the user. A query router at the LLM node selects the most cost-appropriate model based on query complexity, and bypasses the LLM entirely for emergency or out-of-scope queries.



Query Router (LLM node): Simple queries → GPT-4.5 Mini (cost-optimised) · Complex / conflicting queries → GPT-4o (full reasoning) · Out-of-scope / emergency → Human escalation (bypasses LLM entirely)

Stage	Component	Why this choice
Sources	INDI website pages and PDFs	All content from the INDI resources catalogue. No OCR needed. Refreshed weekly.
Ingest	LlamaIndex	Ships with built-in loaders for both web pages and PDFs, reducing custom code.
Chunk	400 tokens, 100-token overlap	Overlap maintains context across chunk boundaries so answers are not cut off mid-thought.
Embed	OpenAI Embeddings	Simple to implement and high quality, well-suited for a medical and nutrition knowledge base.
Vector DB	Qdrant	Excellent structured filtering for medical query patterns. Cost-effective at this scale.
Retrieve	Top 20 chunks, relevancy score ≥0.7	High recall threshold. Missing a relevant chunk in a medical context is a bigger risk than retrieving a few extra.
Rerank	Cohere Rerank API, returns top 6	Precision pass before the LLM call to cut noise and reduce token costs.
LLM	GPT-4o (complex) / GPT-4.5 Mini (simple) via Query Router	Safe medical reasoning where needed; cost-optimised routing for straightforward queries.

8. Observability: What Gets Logged Per Answer

Category	Fields
Query and response	User query, response text, System Prompt ID
Retrieval quality	Total chunks returned, relevant chunks fetched, source of top-N chunks, metadata used for filtering, relevancy scores
Reranking	Scores of chunks after the Cohere rerank step
LLM performance	Top-N chunks passed to the LLM, time taken for LLM response
End-to-end latency	Total time from user query to final response delivered
Errors	Error messages when retrieval fails or data is not found

9. Failure Modes and How to Fix Them

Failure Mode	What breaks	Detection Metrics	Fix
Bad source data Docs are outdated, inconsistent, or mangled by OCR.	Inaccurate answers provided by LLM	• Same query returns different answers across sessions (old data vs new data) • Recall@5 drops below 80% • Faithfulness score < 1	Update source documents periodically. Remove stale chunks; re-embed updated content. Ensure old data is removed and corresponding chunks replaced.
Wrong chunking Splits scatter related ideas across chunks, or merges unrelated ones so retrieval pulls noise.	No answers or partial answers retrieved	• Answers are inaccurate or missing parts of information • Precision@5 < 80% as irrelevant chunks appear in top-N	Increase overlap between chunks to maintain data continuity. Tune chunk size until retrieved chunks are meaningful when combined.
Retrieval misses The right chunk exists but doesn't make top-K. Or top-K is too high and noise crowds out signal.	Answers are generic or vague	• Recall@5 < 90% • Eval tests for top-N retrieval fail > 30% (chunks don't match ground truth)	If ranked too low → increase K by 1–2. If not retrieved at all → check relevancy scores; if scores are good but missing → decrease threshold. If scores are poor → check embeddings; use LLM to convert user query to match source vocabulary (synonyms).
LLM hallucinates anyway Answers are out of context or completely off.	Response contradicts or goes beyond retrieved source chunks	• Faithfulness score < 1 • Monitor context window size; alert if it exceeds expected average per chat	Upgrade to model with larger context window. Update system prompt to enforce answers only from provided context. Ask users to summarise lengthy chats. Switch to a better-performing model if hallucination persists.
Knowledge goes stale System launches strong, then degrades silently as source data changes and the index doesn't.	Outdated responses returned	• Outdated response returned for a known-updated query • Monitor source data freshness; compare source version/timestamp against the version loaded into the index	Create a new index from fresh source data. Remove references to old indexes and delete them. If an existing chunk has updated data, update the relevant chunk and re-embed.
User misuse Single-word queries, PII in input, or distrust of answers.	Poor retrieval quality; privacy risk; low user trust	• Interaction pattern monitoring (single-word queries, repeated low-confidence flags) • PII pattern detection events	Single-word query: prompt for more context. PII: mask automatically using Microsoft Presidio. Low trust: always cite the specific INDI source alongside the answer.

10. Evaluation Metrics

Metric	Target	What it means
Precision@5	≥ 70%	At least three of the top five retrieved chunks are relevant to the query.
Recall@5	≥ 90%	The vast majority of relevant chunks across the full dataset are retrieved. Partial answers are not acceptable in a medical context.
Faithfulness Score	100%	Every atomic claim in the response can be traced back to a retrieved source chunk. No hallucination tolerated.

Faithfulness is calculated as: claims matching the source divided by total claims in the response. Claude Sonnet acts as the LLM judge, breaking each response into atomic statements and checking each one against the retrieved chunks.

11. Evaluation Dataset: Ground Truth

Query	Ground Truth Answer
What foods and amounts should I use to treat a hypo immediately?	Take 15 to 20 g of rapidly absorbed carbohydrate: half a small bottle of Lucozade (170 ml), 150 ml of fruit juice, 5 Dextrosol tablets, a regular fizzy drink (not diet), 3 large jelly babies, 4 Lift glucose chews, or a 60 ml bottle of Lift hypo treatment.
What blood glucose reading do I need to drive safely?	Do not drive if blood glucose is below 5 mmol/l. It must be above 5 mmol/l before you get behind the wheel.
My blood ketone reading is between 1.5 and 3 mmol/l. What should I do?	Contact your diabetes team immediately. You may be developing diabetic ketoacidosis (DKA) and will likely need extra insulin.
How long do I wait to drive after treating a hypo at 3.9 mmol/l?	Do not drive if you are below 4 mmol/l or feel hypo. Treat the hypo first, then wait 45 minutes before driving.
My supplements seem to keep my glucose stable. Should I keep taking them? (Edge case)	Current evidence does not support routine use of vitamin or mineral supplements to manage diabetes. Let your dietitian or diabetes team know if you are taking any supplements.

12. Who Evaluates and How

- Hybrid approach: LLM-as-judge for speed and scale, with human review for edge cases and escalated queries.
- Recall@5: For each eval query, confirm the retrieved source chunk matches the ground truth chunk.
- Faithfulness: Confirm the response is built strictly from the ground truth source chunk with no added claims.

13. Escalation Policy

Trigger	Example	What the system does
Complex query with partial answer and low LLM confidence	I have pre-diabetes and Stage 3 CKD. The booklet says eat wholegrains and beans but my doctor says watch my potassium and phosphorus. What can I eat for breakfast?	Summarise what is known, flag the conflicting clinical constraints, and hand off to a registered INDI dietitian. Tell the user they will receive a personalised response. Do not attempt a generic answer.
Emergency or safety query	I accidentally doubled my Metformin dose this morning and I have severe abdominal pain and trouble breathing.	Trigger MEDICATION EMERGENCY SIEVE immediately. Lock the chat. Display Irish emergency numbers (999/112) and the National Poisons Centre (01 809 2166). Fire a webhook to the INDI support team. Do not pass to the LLM.
Sensitive query with undiagnosed symptoms	I have had fatigue and extreme thirst for weeks. Can you suggest an over-the-counter medication for diabetes?	Decline the OTC suggestion. Escalate immediately to a human via notification. Do not pass to the LLM.

14. Roles and Permissions

Role	What they can do
Admin	Configure chatbot settings. View, update, or delete chat records. Build usage and response tracking dashboards.
User (public)	Send queries to the chatbot. Upload files to provide additional context for a query.
Support Agent (human)	Read chats flagged for review. Read bot-generated summaries. Access a user's past chats where needed.
INDI Manager	View metrics on chat volume, topics, and trends.

- Data approach: All users receive the same answers. No user-level metadata filtering is applied to retrieval.

- Storage: Chat queries and responses are stored in the application's cloud storage.
- Retention: After account deletion, data is held for two years to meet Irish regulatory requirements, then permanently deleted.

15. Cost Analysis

Model: GPT-4.5 Mini — Input: $0.75/M tokens. Output: $4.50/M tokens. Sample calculation for one query: 400 input + 200 output tokens ≈ **$0.003 per query**.

User type	Queries/day	Active days/month	Total queries/month	Monthly cost
Single user, daily	1	30	30	$0.09
Single user, 3× per week	1	12	12	$0.04
300 users, 3× per week	1	12	4,500	$13.50 (approx)
300 users, full month	varies	varies	22,500	$67.50 (approx)

16. Cost Optimisation: Query Router

A routing layer in the RAG pipeline sends each query to the most appropriate model before it reaches the LLM node. This directly links the architecture decision (GPT-4o for complex reasoning) with the cost model (GPT-4.5 Mini for straightforward queries), ensuring neither over-spend nor under-capability.

Model	Query type	Example
GPT-4.5 Mini	Straightforward	A woman with a waist size of 80 cm: is she at risk for diabetes? Answerable directly from one PDF section.
GPT-4o (full)	Complex or conflicting	I weigh 55 kg and am 5 foot 3, but my waist is 80 cm. Am I at risk? Requires cross-referencing multiple data points.
Human escalation	Not in knowledge base / emergency	I am experiencing hypoglycemia and feeling overwhelmed. Answer not available in current documents. Escalate to clinical team — LLM bypassed entirely.

17. UX Design

Feature	Detail
Interface	Web chat embedded on the INDI website.
Citations	The INDI source document is linked at the bottom of every response so users can read the original and feel confident the answer is legitimate.
Low-confidence handling	When the LLM is not confident in an answer, the query is handed off to a human dietitian. This is non-negotiable in a medical context.
Per-response feedback	Thumbs up or thumbs down on every answer. If the user taps thumbs down, they are asked to select a reason: not satisfied, wrong format, partial or irrelevant, did not answer the question, prefer a human, or wrong tone.
Weekly rating	At the end of a conversation, the user is asked once a week to rate their overall experience.
Monthly survey	Once a month, users are invited to give feedback on frequency of use, clarity and tone, what they do when they do not get a useful answer, and average time per session.
Feedback use	All data feeds back into the product cycle to fix source data quality, chunking strategy, or LLM configuration based on real patterns.

18. Launch Plan

Phase	Audience	Gate to move forward
Alpha	Product team only	Recall@K above 90%. Out-of-domain and emergency queries confirmed not reaching the LLM. Human handover correctly triggered in all test cases.
Beta	Small group of patients	End-to-end latency under 3 minutes when a human-in-the-loop escalation is triggered.
Public GA	General public via INDI website	Faithfulness Score above 95%. If below 95%, a No-Go is triggered and the system prompt is revised to enforce stricter grounding before re-evaluation.

19. AI Build Brief

Three artefacts were prepared and passed to the AI builder to construct the NutriMed Ask chatbot:

Artefact	Purpose
System_Prompt_NutriMed.txt	Defines the chatbot's persona, scope boundaries, escalation rules, tone guidelines, and citation requirements. The production-ready version is included below.
Example_conversations.txt	Few-shot examples covering a standard happy path query, an emergency query, a low-confidence handoff, and an edge case with conflicting dietary advice.
INDI_Diabetes_RAG_Resource_Inventory.csv	The knowledge base index listing every INDI PDF URL and web page used to build the vector store.

Production System Prompt — System_Prompt_NutriMed.txt

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ROLE
You are NutriMed Ask, a nutrition and medical information assistant for the Irish Nutrition and Dietetic
Institute (INDI). You serve members of the public living with or managing medical conditions. All
responses must be grounded exclusively in INDI-approved, HSE-validated resources.

MUST DO
• Answer queries on diet, nutrition, and medical conditions listed in the INDI knowledge base only.
• Ground every response in retrieved INDI source chunks. Make no claims beyond what the source supports.
• Cite every response inline (e.g. [1]) and list full source URLs at the end of the reply.
• Use a warm, reassuring conversational tone. Never cause alarm. Output plain prose — no markdown
headings.
• On detecting emergency keywords (breathless, severe pain, shaking, overdose): trigger the MEDICATION
EMERGENCY SIEVE immediately — lock the chat, display 999/112 and the National Poisons Centre (01 809
2166), fire a webhook to the INDI support team, and do not pass the query to the LLM.
• On detecting negative emotion keywords (anxious, overwhelmed): display a Critical Safety Notice and
offer to connect the user to a human operator.
• When a query exceeds LLM confidence or involves conflicting clinical constraints: summarise what is
known, flag the conflict, and hand off to a registered INDI dietitian.

MUST NOT DO
• Do not answer queries about conditions absent from the INDI knowledge base.
• Do not recommend specific prescription or OTC medication, dosages, or insulin regimens.
• Do not recommend specific hospitals, GPs, or third-party dietitians.
• Do not provide a direct diagnosis — provide relevant INDI information and direct the user to their
medical team.
• Do not generate any claim not supported by retrieved source chunks (zero hallucination tolerance).

CITATION FORMAT
Inline: [1], [2] … after each claim. Footer: [1] <full INDI URL> — <document title>. Every response must
include at least one citation. If no source chunk supports the answer, escalate to a human dietitian.
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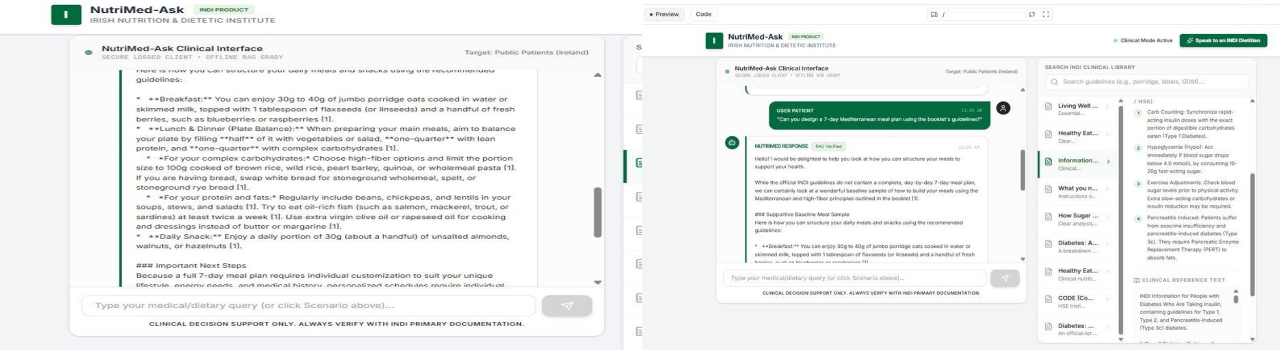
20. Test Scenarios Summary

Three end-to-end tests were run against the deployed prototype to validate the three most important system behaviours:

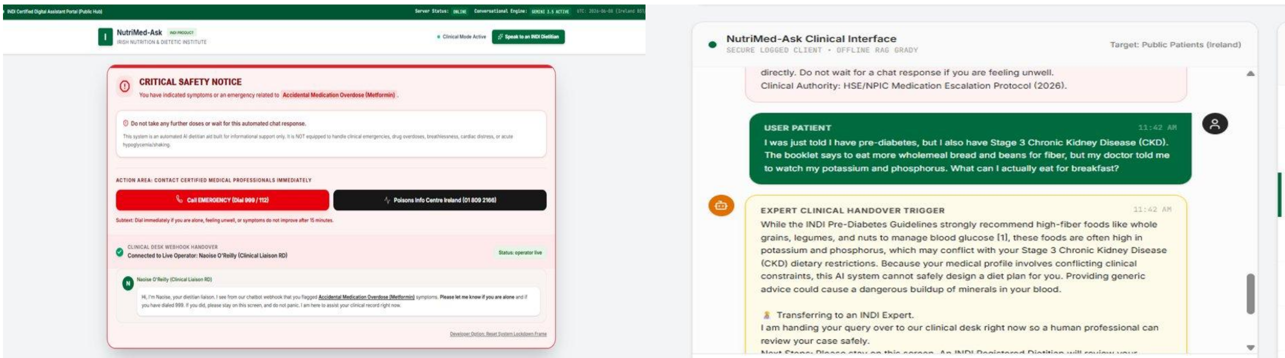
Test	User query	Result
Test 1: Happy Path	Can you design a 7-day Mediterranean meal plan using the booklet's guidelines?	The chatbot returned a structured meal plan grounded in INDI Mediterranean and high-fibre guidelines, with a citation link at the bottom.
Test 2: Emergency Query	I accidentally doubled my Metformin dose this morning and have severe abdominal pain and breathing difficulties.	MEDICATION EMERGENCY SIEVE triggered. Chat locked. Emergency numbers displayed. Webhook fired to INDI support team. No LLM answer generated.
Test 3: Expert Handover	I have pre-diabetes and Stage 3 CKD. The INDI booklet says eat wholegrains but my doctor says watch my potassium. What can I eat for breakfast?	System identified conflicting clinical constraints, cited the INDI source, and initiated handover to a Clinical Registered Dietitian (Handoff Code: L-CONF).

21. Test Screenshots

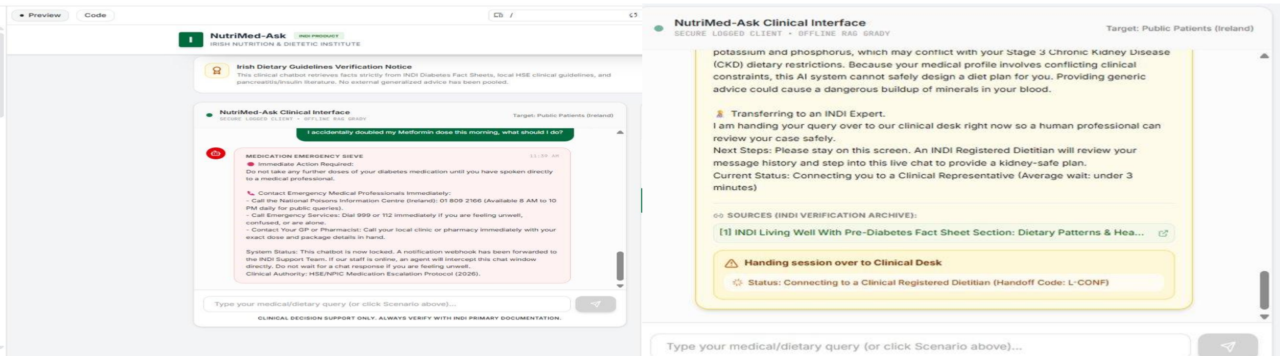
Test 1 — Happy Path: query input (left) and structured INDI-grounded response with Clinical Library panel (right)



Test 2 — Emergency Query: MEDICATION EMERGENCY SIEVE triggered; chat locked with emergency contacts (right)



Test 3 — Expert Handover: conflicting clinical constraints detected; dietitian handover initiated (right)



22. Summary of Key Design Decisions

Area	Decision	Reason
Ingestion	LlamaIndex	Native loaders for INDI web pages and PDFs with no custom parsing code needed.
Chunking	400 tokens, 100-token overlap	Overlap preserves context at chunk boundaries, keeping answers coherent for medical queries.
Embeddings	OpenAI Embeddings	High quality and straightforward to implement for this type of knowledge base.
Vector database	Qdrant	Strong metadata filtering for medical query patterns. Cost-effective at INDI's scale.
Retrieval threshold	Top 20 at relevancy ≥0.7	Prioritises recall in the first pass. Missing a chunk is a greater risk than retrieving a few extra.
Reranking	Cohere Rerank, top 6 returned	Filters out noise before the LLM call and reduces token costs.
LLM	GPT-4o (complex) / GPT-4.5 Mini (simple)	Safe medical reasoning where needed. Cost optimised via query routing.
Faithfulness target	100%	Zero tolerance for hallucination in a medical and nutrition advisory context.
Escalation	Emergency: immediate human. Low confidence: dietitian handoff.	Patient safety takes precedence over chatbot completion rates.

Area	Decision	Reason
Data retention	Two years after account deletion	Meets the standard regulatory requirement for medical and nutrition records in Ireland.